



CEBU INSTITUTE OF TECHNOLOGY
UNIVERSITY

IT342 - G1

SYSTEMS INTEGRATION

AND ARCHITECTURE 1

FUNCTIONAL REQUIREMENTS

SPECIFICATION (FRS)

Project Title: Mini App – User Registration & Authentication

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1. Introduction

1.1. Purpose

The purpose of this system is to allow users to create accounts, log in securely, view their dashboard/profile, and log out. This document is intended for students, developers, and instructors to understand how the system works.

1.2. Scope

The system will handle user registration, login, profile viewing, and logout. It focuses only on basic account management and does not include advanced features like payments or messaging.

1.3. Definitions, Acronyms, and Abbreviations

- **User** – A person using the system
- **Register** – Creating a new account
- **Login** – Accessing an account using credentials
- **Logout** – Exiting the account securely
- **Dashboard** – Main page after login
- **Credentials** – Username and password

2. Overall Description

2.1. System Perspective

The system is a standalone web or mobile application that manages user accounts. It can be part of a larger system like a school portal or business app.

2.2. User Classes and Characteristics

Guest User

- Not logged in
- Can only register or login

Registered User

- Has an account
- Can login, view dashboard, and logout

2.3. Operating Environment

- Smartphone or computer
- Web browser or mobile app
- Internet connection
- Database server for storing accounts

2.4. Assumptions and Dependencies

- Users have internet access
- Users provide correct information
- System depends on a database to store user data

3. System Features and Functional Requirements

Describe each major feature of the system and its functional requirements.

3.1. Feature 1: User Registration

Description: Allows new users to create an account.

Functional Requirements:

- User enters name, email, and password
- System validates input
- System saves data in database
- System confirms successful registration

3.2. Feature 2: User Login

Description: Allows registered users to access their account.

Functional Requirements:

- User enters email and password
- System verifies credentials
- System grants access if correct
- System shows error if incorrect

3.3. Feature 3: Dashboard/Profile Viewing

Description: Allows logged-in users to see their account details.

Functional Requirements:

- Display user info
- Show welcome message
- Restrict access if not logged in

3.4. Feature 4: Logout

Description: Allows users to securely exit their account.

Functional Requirements:

- User clicks logout
- System ends session
- User returns to login page
- Protected pages become inaccessible

4. Non-Functional Requirements

Performance: System should respond quickly

Security: Passwords must be protected

Usability: Simple and easy interface

Reliability: System should work consistently

Availability: Accessible anytime with internet

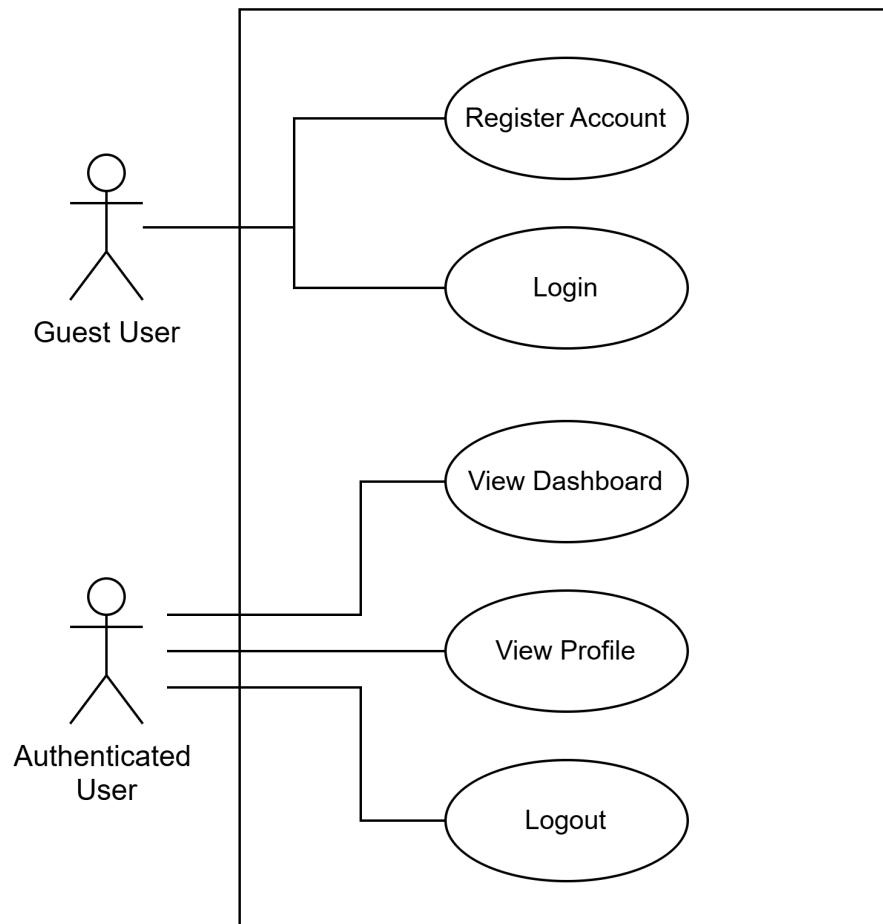
5. System Models (Diagrams)

Insert the necessary diagrams for the system:

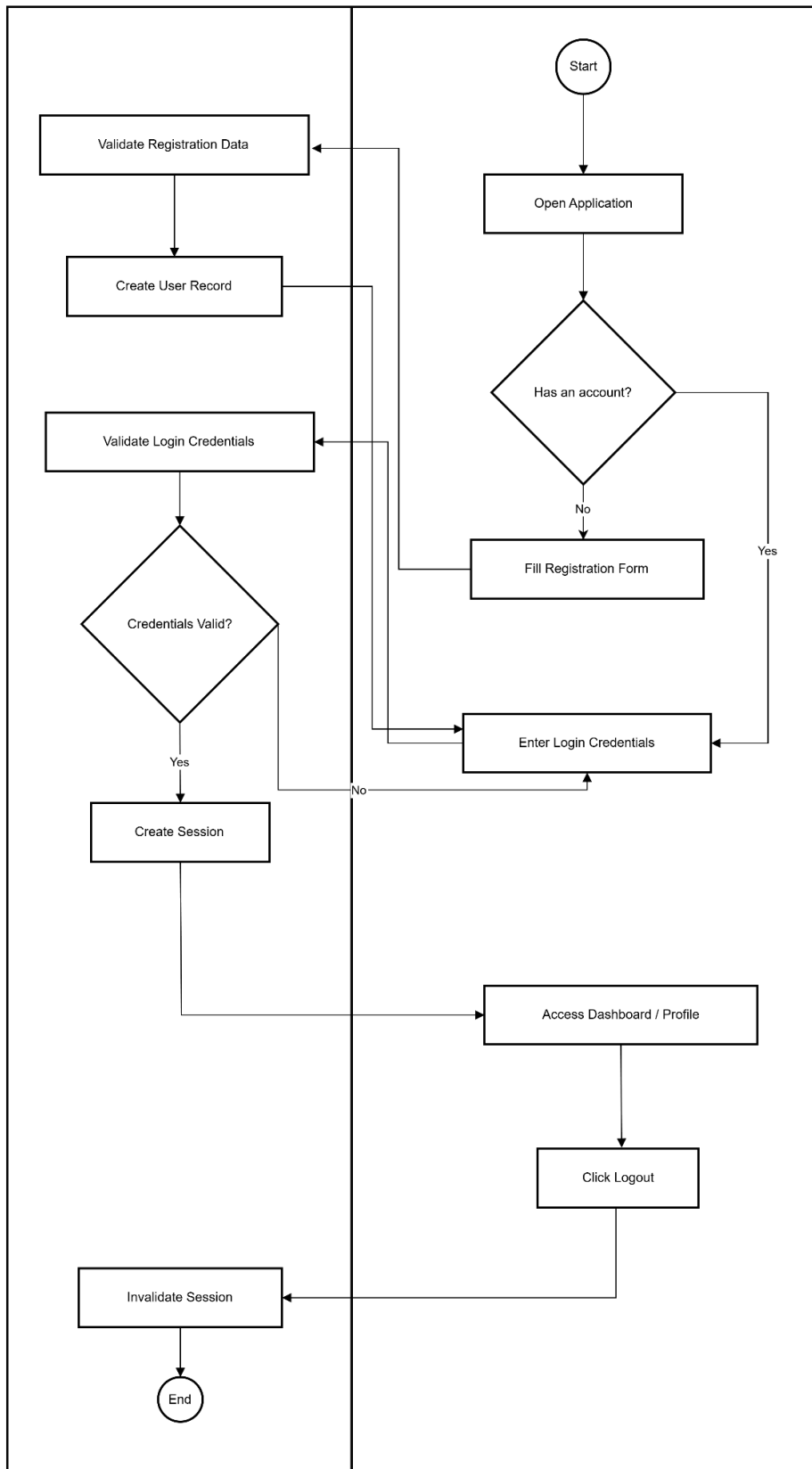
5.1. ERD

Users	
PK	<u>user_id</u>
	username
	first_name
	last_name
	email
	role
	password_hash

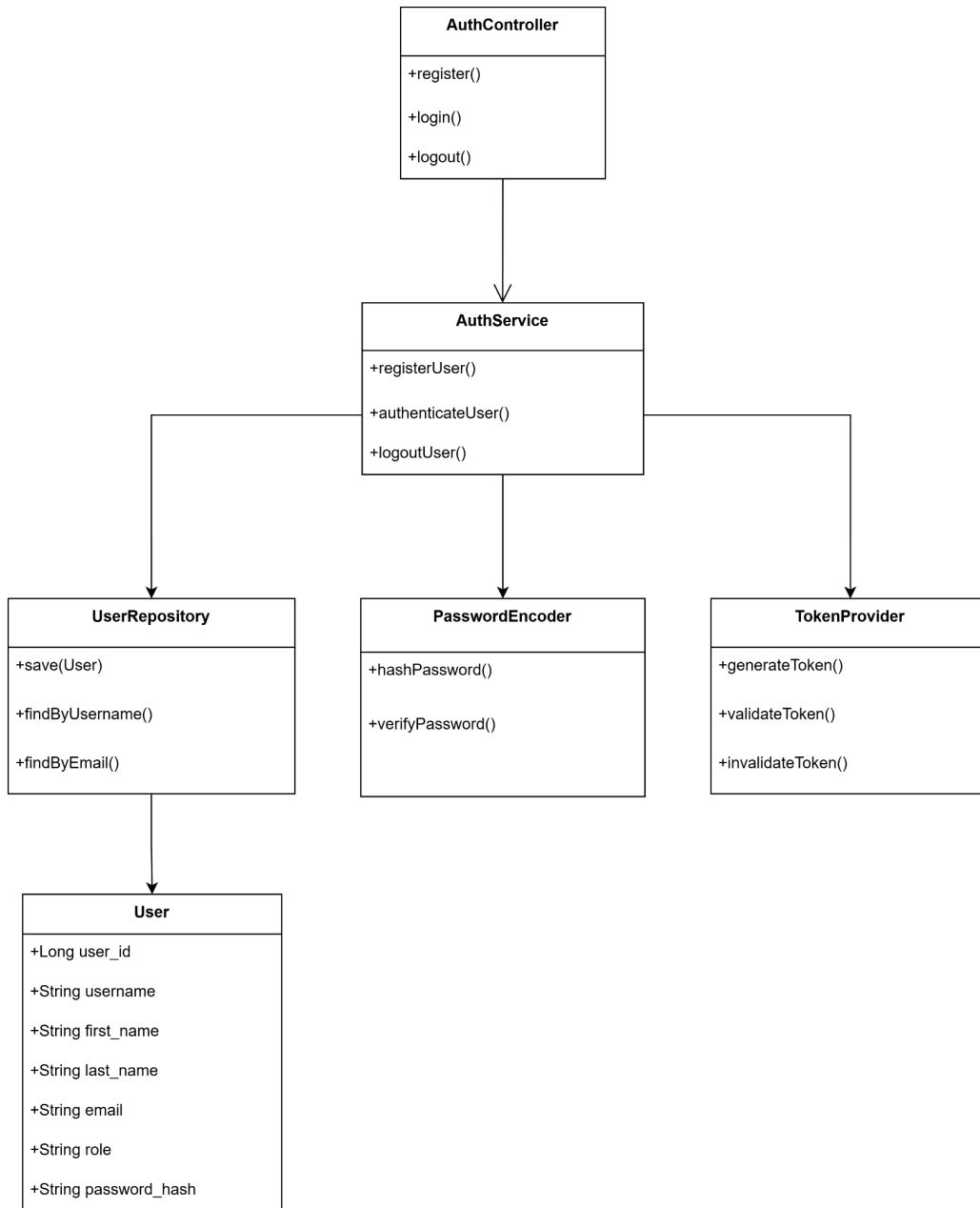
5.2. Use Case Diagram



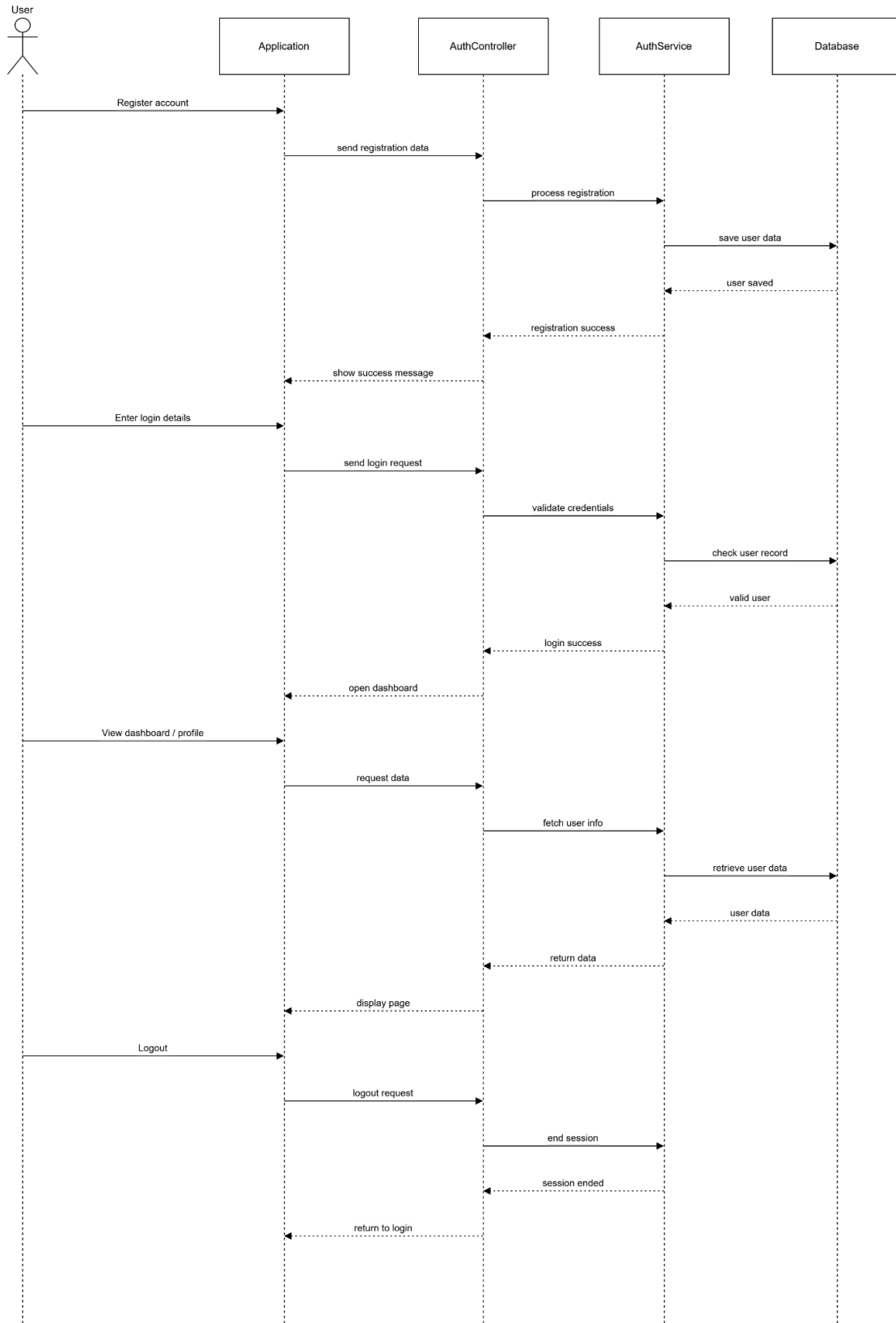
5.3. Activity Diagram



5.4. Class Diagram



5.5. Sequence Diagram



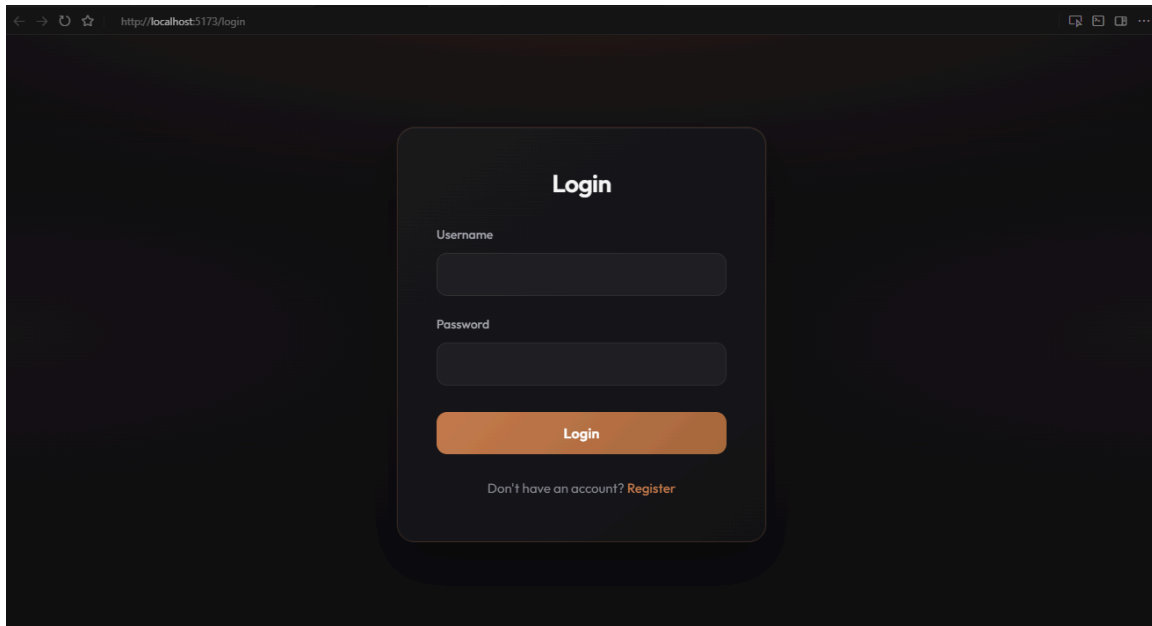
6. Appendices

Appendix A - References

- OWASP Security Guidelines for Authentication
- UML Diagramming Standards

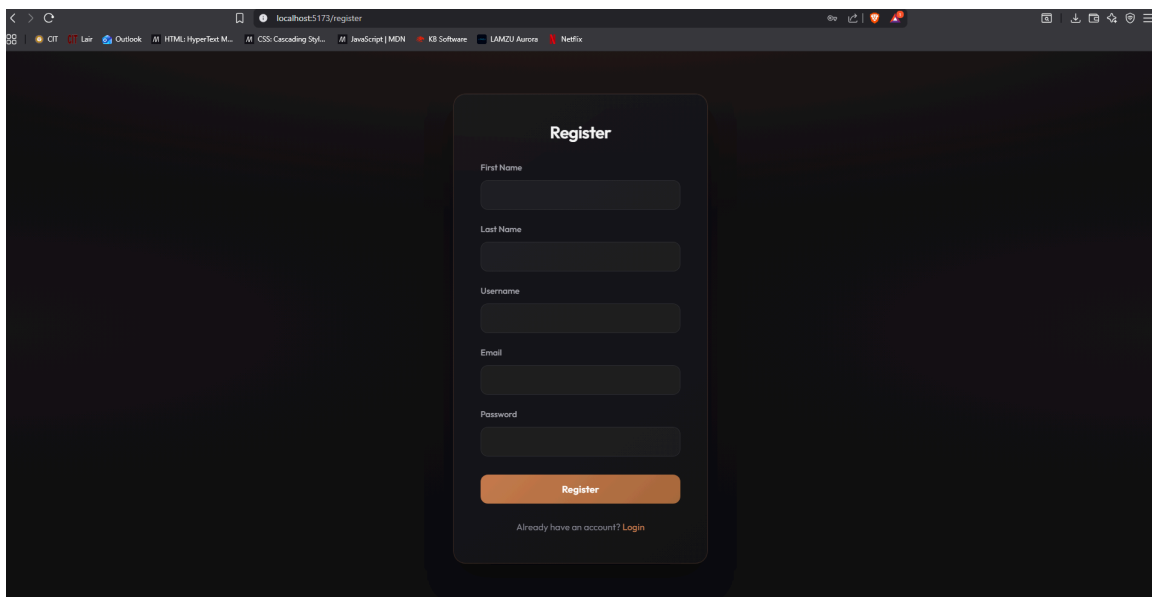
Appendix B - Screenshots

Login Page for Web:



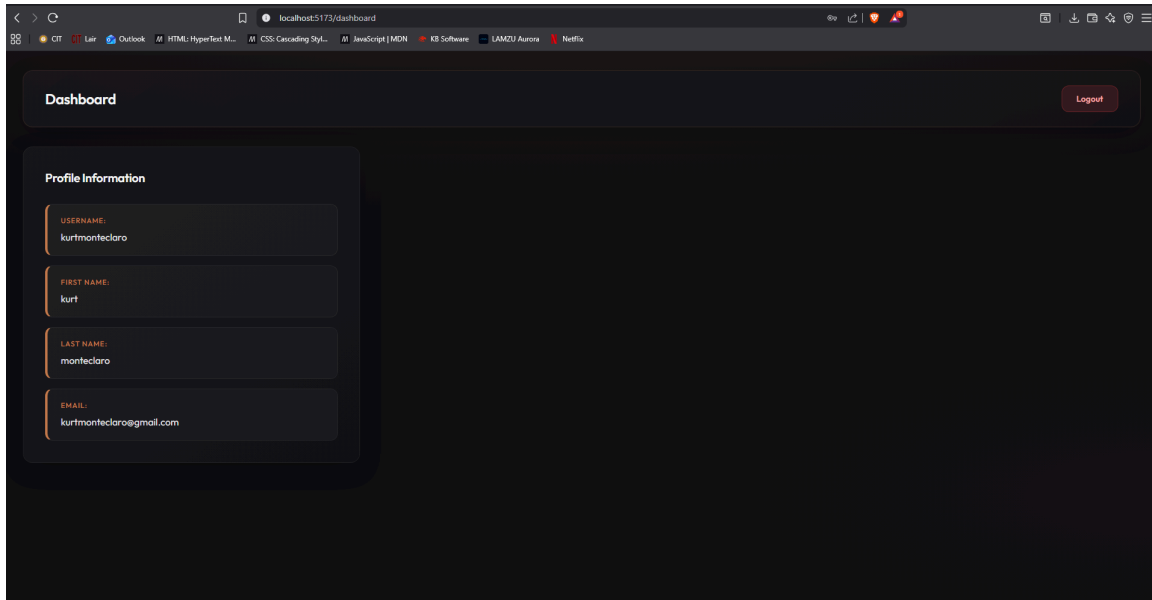
The screenshot shows a web browser window with the address bar displaying `http://localhost:5173/login`. The page features a dark background with a central light gray box containing the 'Login' form. The form includes fields for 'Username' and 'Password', a 'Login' button, and a link to 'Register'.

Register Page for Web:

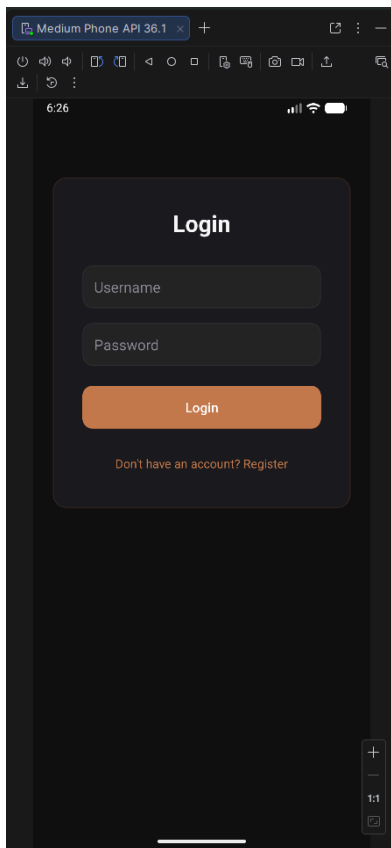


The screenshot shows a web browser window with the address bar displaying `localhost:5173/register`. The page features a dark background with a central light gray box containing the 'Register' form. The form includes fields for 'First Name', 'Last Name', 'Username', 'Email', and 'Password', a 'Register' button, and a link to 'Login'.

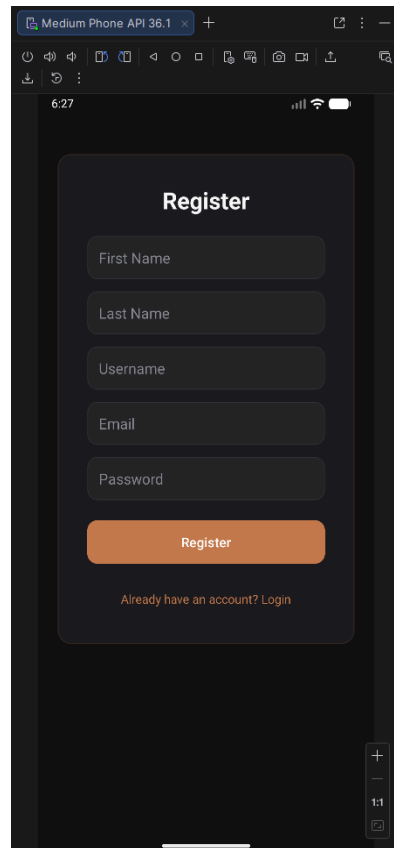
Dashboard Page for Web:



Login Page for Mobile:



Register Page for Mobile:



Dashboard Page for Mobile:

